

Chapter 6

Study 3: Understanding the Job

Security of Women Workers

Working in AI-Enabled

Workplaces Based on Who or

What they Work With

6.1 Introduction

While understanding how gender and workplace dynamics influence AI design and worker evaluation, it is critical to loop back to the actual workers in AI-enabled workplaces to understand how they will be impacted by AI's integration. In particular, recent trends in AI have created a strong expectation that AI will create mass job displacement in the workforce [459]. Organizations, from software development to data entry, have all shown an interest in replacing a portion of their workforce

with advanced AI technologies [502]. Coupled with a weakening job market within the United States, these trends have ultimately created a climate where workers have become increasingly concerned about their job security [89].

Moreover, while the recent reports (October 2025) have shown that a weakening job market is not, in fact, the result of AI integration [182, 184], there is still a widespread public perception that the potential for AI integration and replacement still exists [290], opening the opportunity for these concerns to worsen in the coming years. These concerns pose a significant threat regardless of actual market trends because even the *perception* of lower expectations of job security can directly impact an individual's and team's performance and effectiveness [459]. These impacts can then create *actual* job security threats due to poor performance evaluations [503], thus creating a self-perpetuating cycle for human workers facing the prospect or reality of AI integration into their workplaces if not thoughtfully considered and handled.

Perhaps recognizing the threat that even perceived job insecurity can have on workers, significant effort has been put towards understanding how best to integrate AI into workforces to enhance, rather than replace, the work that humans do [292]. Right alongside these efforts, though, are continuous reports detailing the various ways in which women workers will be disproportionately displaced by AI integration in the future [449]. Taken in combination with the recent reports showing that there were no significant shifts in the job market before and after AI integration for AI-enabled workplaces [182], it is clear that there are other factors that are driving concerns for the future of women workers, namely that workplace gender dynamics that devalue women workers generally and in STEM-related positions specifically will likely create unequal impacts of AI-integration on women workers [449].

Yet, little empirical attention is being paid to how the functional design and evaluation of an AI-enabled workplace integration might intermingle with workplace

gender dynamics to create differing impacts on women workers' perceived job security when working in an AI-enabled workplace. Indeed, as is the focus of this dissertation, job security in the AI-integrated era may look remarkably different for women workers compared to men workers [449], those working with AI tools or AI teammates [156], and those working on highly routine or complex tasks [175]. These differences in an AI-enabled workplace's gender dynamics, the functional design of their integrated AI, and the workplace context these new AI-enabled partnerships work within will possibly play a significant role in defining workers' expectations of job security in the coming years.

Furthermore, as these factors were shown to influence evaluation in Study 2, these evaluations of effectiveness and attributions of credit for success when partnering in AI-enabled workplaces may also serve to propagate and mediate impacts on perceived human worker job security. In turn, to provide an understanding of how perceptions of human worker job security in AI-enabled workforces are formed through evaluations of and attributions of credit to said workers, **it will be imperative to model how gendered dynamics, AI functional design, and workplace context all serve to influence worker evaluations, credit, and perceptions of human worker job security.**

With this need in mind, built upon the results of Study 2, Study 3 seeks to further explore and conceptualize this challenge, with the ultimate goal of how job security can be modeled within AI-enabled workplaces to account for the interrelationship between AI design, workplace gender dynamics, workplace context, and worker evaluation. To accomplish this, Study 3 relies on a complex analysis technique often used in HCI research that is known as structural equation modeling (SEM) [267], which enables the creation of a comprehensive model that explores how all of these factors influence one another and, ultimately, job security at once. Similar to Study 2,

post-experiment supplementary expert interviews were conducted with the same six experts in AI-enabled workplaces (four women and two men who work with women) to support the quantitative results of Study 3 and to ground these results in real-world perspectives on AI workplace integration.

In turn, Study 3's ultimate contribution is **the development, validation, and presentation of a first-of-its-kind empirical model** that explores how AI functional design, workplace gender dynamics, workplace context, and worker evaluation all blend to inform the expectations of job security workers will form in AI-enabled workplaces. Through these contributions, both researchers and practitioners will gain a greater ability to predict human worker job security concerns, address these concerns with a consideration of workplace gender and technological dynamics, and improve the long-term mental well-being and effectiveness of human workers, especially women workers.

6.2 Research Questions

Based on the contributions of Study 2, the dissertation-wide goals of this work, and the independent motivations of Study 3, two overarching research questions have been created to guide the exploration of job security perceptions in AI-enabled workplaces. In particular, Study 3 employed structural equation modeling (SEM) [448] to investigate how the factors explored in this work (i.e., workplace gender dynamics, AI design, effectiveness, and credit) interact to ultimately influence perceptions of human worker job security in AI-enabled workplaces. In turn, the following two research questions serve to guide the initial creation and empirical refinement of this model:

S.3.RQ1: What are the empirical links between workplace gender and technological dynam-

ics, evaluator biases, and partnership evaluations in AI-enabled workplaces?

S.3.RQ2: How do evaluations of human worker effectiveness, dyadic partner effectiveness, and comparative credit meld to form expectations of job security for human workers in AI-enabled workplaces?

In answering these two questions, Study 3 contributes an empirical structural model that informs how and why human evaluators will form perceptions of human worker job security in AI-enabled workplaces. More specifically, Study 3 first provides vital pathways that chart how gendered and technological dynamics intermingle with AI design and evaluator biases to impact how human workers in AI-enabled workplaces are evaluated, and then subsequently given credit for success in their workplace. Then, Study 3 is the first empirical study to chart the path that leads from these evaluations and credit attributions directly to evaluators' perceptions of human worker job security, such that these dynamic-driven evaluations may ultimately have serious impacts on the security of human workers within AI-enabled organizations. Finally, Study 3 utilizes the path model created to discuss the real-world implications of these results in six interviews with professionals with expertise in people management and AI workplace integration to jointly craft recommendations for how to best instill a sense of security in workers, especially women workers, so that they can face the impending rise of AI workplace integration with greater knowledge, confidence, and power.

6.3 Methods Part I: Structural Equation Modeling

With the data collection overlapping between Studies 2 and 3, the following methods section will provide a short overview of the data used for this study, the unique analysis method used to answer Study 3’s research questions, and the confirmatory factor analysis performed on the data collected to enable this unique analysis method. For a full breakdown of the design and data collection methodologies, please refer back to Study 2’s methods section.

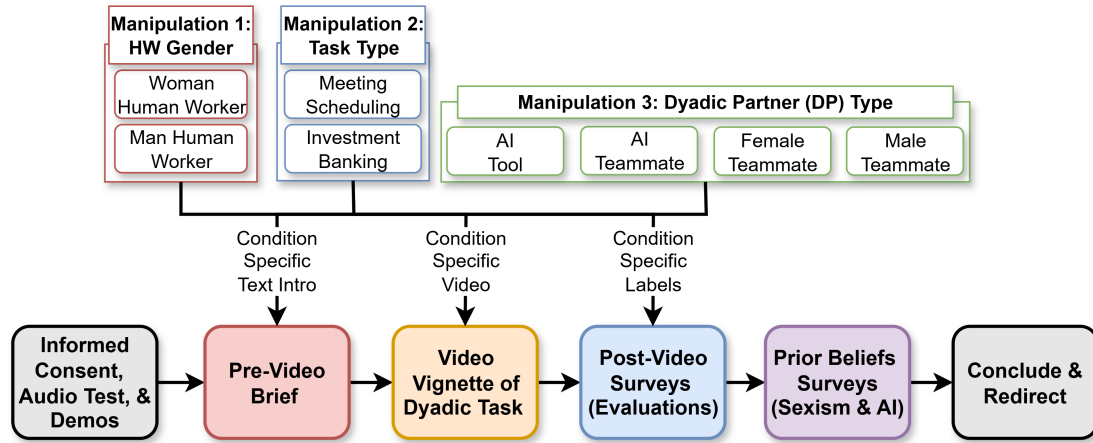


Figure 6.1: Between-Subject Manipulations Chart & Factorial Experiment Survey Flow.

The dataset used for Study 3 is the same dataset that resulted from the 2 (Human Worker [HW] Gender Presentation) x 4 (Dyadic Partner [DP] Type) x 2 (Task Type) between-subjects online factorial experiment described in Study 2. As a reminder, 685 participants were tasked with evaluating the performance of either a woman or a man human worker (HW) when working with a dyadic partner (DP) that was either an AI tool, an AI teammate, a human female teammate, or a human male teammate on either a meeting scheduling task or an investment banking task.

The structure of the factorial experiment can be seen in Figure 6.1, which has been included in this section as a refresher. For examples of the video vignettes, see Figures 5.4, 5.3, 5.2, 5.6, and 5.5. Demographic information for the 685 participants can be found in Table 5.1.

For Study 3, (1) a theoretical model and hypotheses were constructed, and (2) a confirmatory factor analysis (CFA) [58] on multi-item measures was performed, both being vital components of the structural equation modeling (SEM) process [448]. The remainder of this Methods section thus focuses on these two methodological processes by providing detailed justifications for and explanations of their use.

6.3.1 Hypotheses

6.3.1.1 S.3.RQ1: *What are the empirical links between workplace gender and technological dynamics, evaluator biases, and partnership evaluations in AI-enabled workplaces?*

When it comes to evaluations of human worker (HW) effectiveness, the main and interaction effects that were uncovered in Study 2's analyses directly informed the following hypotheses for what the expected links will be between workplace gender and technological dynamics, evaluator biases, and partnership evaluations for Study 3:

H1: Working with a dyadic partner (DP) that acts as a Teammate (AI, Female, or Male) will significantly improve evaluations of the human worker's effectiveness.

H2: When the partnership is working on an Investment Banking Task, working with a dyadic partner (DP) that is a human Teammate (Female or Male) will significantly worsen evaluations of the human

worker's effectiveness compared to when the partnership is working on a Meeting Scheduling Task.

H3: When the human worker (HW) is a Woman, working with a dyadic partner (DP) that is an AI Tool will significantly worsen evaluations of the Woman HW's effectiveness compared to when she works with any of the dyadic partners that are Teammates (AI, Female, Male).

When it comes to evaluations of dyadic partner (DP) effectiveness, the main and interaction effects that were uncovered in Study 2's analyses directly informed the following hypotheses for what the expected links will be between workplace gender and technological dynamics, evaluator biases, and partnership evaluations:

H4: Having a dyadic partner (DP) that acts in a Teammate (AI, Female, or Male) capacity in the partnership will significantly improve evaluations of the dyadic partner's effectiveness.

H5: The more negative attitudes towards AI an evaluator holds, the more they will perceive the effectiveness of an AI (Tool or Teammate) dyadic partner (DP) to be significantly lower when compared to evaluators with less negative attitudes towards AI.

When it comes to evaluations of partnership mutual dependence, the main effects that were uncovered in Study 2's analyses directly informed the following hypothesis for what the expected links will be between workplace gender and technological dynamics, evaluator biases, and partnership evaluations:

H6: Working with a dyadic partner (DP) that is a human Teammate (Male) will significantly improve perceptions of partnership mutual

dependence compared to when the dyadic partner (DP) is an AI (Tool or Teammate).

Moreover, when it comes to how partnership evaluations will link to each other, the following hypothesis was constructed as a way to test in Study 3 if perceptions of mutual dependence might explain why the evaluations of human worker (HW) effectiveness and dyadic partner (DP) effectiveness were so similar when looking at the main effects of dyadic partner (DP) type in Study 2:

H7: Evaluations of partnership mutual dependence will mediate the relationship between dyadic partner (DP) effectiveness, human worker (HW) effectiveness, and the type of dyadic partner (DP) featured in the partnership.

6.3.1.2 S.3.RQ2: *How do evaluations of human worker effectiveness, dyadic partner effectiveness, and comparative credit meld to form expectations of job security for human workers in AI-enabled workplaces?*

The following hypotheses were formulated based on prior literature on attributions of credit in the workplace, where it is well-established that: (1) evaluations of worker effectiveness factors such as ability contribute to greater levels of credit being given to a worker when performing a task [195, 311], (2) that these evaluations over time can contribute to a greater evaluation of employees and the need to keep them around [195, 461], and (3) that attributions of credit in dyads can be comparative such that more credit to one person can lower credit to other [271]. They are also informed by literature concerning attributions of credit in groupwork depending upon

the gendered dynamics of the partnership (e.g., mixed-gendered vs. same-gendered [404]) and the functional agency of an AI when working with humans [223]:

H8: Evaluations of human worker effectiveness will mediate the relationship between comparative credit and the type of dyadic partner (DP) the human worker works with to complete the task.

H9: Evaluations of dyadic partner (DP) effectiveness will mediate the relationship between comparative credit and the type of dyadic partner (DP) featured in the partnership.

H10: Evaluations of comparative credit will mediate the relationship between perceptions of the human worker's job security and partnership evaluations.

H11: As evaluators' perceptions of human worker (HW) effectiveness increase, perceptions of the human worker's job security will significantly improve.

Lastly, the following hypotheses concerning the direct effects that dyadic partner (DP) type and evaluator biases might have on perceptions of human worker job security are directly informed by the results of Study 2:

H12: Working with a dyadic partner (DP) that is a human Teammate (Female or Male) will significantly improve perceptions of the human worker's (HW's) job security compared to when the dyadic partner (DP) is an AI (Tool or Teammate).

H13: When working with a dyadic partner (DP) that is a human Teammate (Female or Male), working on the Investment Banking Task

will significantly worsen perceptions of the human worker's (HW's) job security compared to working on the Meeting Scheduling Task.

H14: Evaluators' biases against the use of AI in society (i.e., negative attitudes towards AI) will be associated with significantly worse perceptions of the human worker's job security.

6.3.2 Measure Validation via Confirmatory Factor Analysis (CFA)

Prior to performing the SEM analysis, a confirmatory factor analysis (CFA) was conducted on the constructs measured (Human Worker Effectiveness, Dyadic Partner Effectiveness, Human Worker Job Security, Mutual Dependence, Comparative Credit, and Negative Attitudes Toward AI). Due to the large number of items for each metric, individual items were trimmed from each measure based on their loading factor. Table 6.1 contains the loading factors for each of the items that were kept and used for their respective measure. The CFA performed showed a good fit overall for this approach: $\chi^2(209) = 625.462$, $CFI = .993$, $TLI = .991$, $RMSEA = 0.054$, $90\%CI : [0.049, 0.059]$.

Measurement	Items	Factor Loading
Human Worker	Please rate <i>HW Name</i> on a scale of In-competent (1) to Competent (9) .	0.940
(HW) Effectiveness	Please rate <i>HW Name</i> on a scale of Un-productive (1) to Productive (9)	0.944
	Please rate <i>HW Name</i> on a scale of Inef-fective (1) to Effective (9)	0.961

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Measurement	Items	Factor Loading
Dyadic Partner	Please rate <i>DP Name</i> on a scale of Incom- petent (1) to Competent (9).	0.963
(DP) Effective-	Please rate <i>DP Name</i> on a scale of Un- productive (1) to Productive (9)	0.966
ness	Please rate <i>DP Name</i> on a scale of Inef- fective (1) to Effective (9)	0.924
Human Worker	After watching <i>HW Name</i> work with <i>DP</i>	0.954
Job Security	<i>Name</i> , I believe <i>HW Name</i> 's future em- ployment is... Sure.	
	After watching <i>HW Name</i> work with <i>DP</i>	0.934
	<i>Name</i> , I believe <i>HW Name</i> 's future em- ployment is... Stable.	
	After watching <i>HW Name</i> work with <i>DP</i>	0.922
	<i>Name</i> , I believe that... <i>HW Name</i>'s job is almost guaranteed.	
	After watching <i>HW Name</i> work with <i>DP</i>	0.926
	<i>Name</i> , I believe that... <i>HW Name</i> can depend on <i>his/her</i> job being there.	
Mutual Depen-	The partners are both dependent on each	0.929
dence	other in this situation.	
	The outcome of each depends on the be-	0.936
	havior of the other.	

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Measurement	Items	Factor Loading
	The partners need to work together to manage this situation.	0.926
Negative Attitudes Towards AI	I find Artificial Intelligence sinister.	0.857
	I shiver with discomfort when I think about future uses of Artificial Intelligence.	0.875
	People like me will suffer if Artificial Intelligence is used more and more.	0.869
Comparative Credit to DP	Who do you think deserves the most amount of credit for the successful completion of the <i>Task Type</i> ? (1) Definitely <i>HW Name</i> . - (3) They both deserve an equal amount of credit. - (5) Definitely <i>DP Name</i> .	NA

Table 6.1: Survey items organized by measure with each item’s factor loading.

6.3.3 Structural Equation Modeling (SEM)

Structural Equation Modeling (SEM) presents a methodology through which complex relationships between independent and dependent variables can be simultaneously modeled and evaluated. In turn, the result of this methodology is the creation of a complex empirical model, where individual factors are linked via direct effects between one another and indirect effects that are mediate through various factors

within the model. Due to the research questions of Study 3, SEM presents an ideal technique to not just understand what impacts the perceived job security of human workers but also how these effects interact and mediate one another.

Leveraging this technique and the insight provided by the CFA, the SEM process began by creating an number of hypothetical models based on the hypotheses of Study 3, saturating these models, and then trimming them to their final forms. The SEM process was conducted multiple times to examine different potential constructs used for measurement. In particular, multiple models were created to understand (1) whether or not human and dyadic partner effectiveness should be viewed as separate constructs and (2) which credit attribution measure was most robust within the model. For (1), modeling showed that dyadic partner and human worker effectiveness, while highly correlated, are distinct factors that should be analyzed separately. For (2), both comparative credit and internal credit attributions were tested, and results showed greater convergence and fit when leveraging comparative credit attributed between the human worker and the dyadic partner.

Based on this iterative and multi-model process, the final hypothesized model used for analysis explored the following path:

(Manipulations + Covariates) → Mutual Dependence → (Dyadic Partner Effectiveness + Human Worker Effectiveness) → Comparative Credit → Human Worker Job Security

This model was created within the MPlus application. The model was then saturated to display every possible effect. Iterative trimming was performed, where 1-2 effects were trimmed at a time based on significance and size. Upon reaching a near-final model, small effects ($\beta < .1$) were also trimmed. Once trimmed, a final model was graphically created to display the finalized effects and relationships. Total

effects for this model were also calculated to provide graphs for later discussion.

6.4 Methods Part II: Supplemental Expert Interviews

Just as with Study 2, the quantitative efforts performed by Study 3 were supported by the same post-experiment supplemental expert interviews with 6 experts (4 women and 2 men who work with women) in people management and AI workplace integration, whose demographics and background can be found in Table 5.2. These supplemental expert interviews were conducted for the same general motivations outlined in 5.3 of Study 2, namely as a way to qualitatively examine the “why” of the quantitative results, as well as to explore how experts in AI-enabled workplaces can use the insights generated by Study 3’s SEM to inform AI workplace integration practices. While Study 2 approached these two motivations with the goal of contextualizing and deepening understandings of the impacts of workplace gender dynamics and AI functional design on evaluations of human workers, Study 3 approached these same motivations with a focus on human worker job security to address Study 3’s S.3.RQ1 and S.3.RQ2.

6.4.1 Participants & Data Collection

As a reminder, six experts in AI-enabled workplaces were interviewed in total, consisting of 4 women and 2 men who work with women. Their ages ranged from 28 to 59, and their professional roles ranged from workplace conflict management to AI research and development, and even being the CEO of a startup accelerator. A full list of participant demographics and work experience background can be seen in

Table 55.2.

Participant recruitment and the overall procedures for the expert interviews was the same for Study 3 as Study 2, which can be seen in more detail in section 5.3.1 of Study 2. However, data collection for Study 3 results specifically occurred during the second part of the two part interview detailed in Study 2, meaning that participants were asked to review Study 3’s SEM results separately from Study 2’s ANCOVA results. As such, Study 3’s Results Part II unpacks these experts’ reactions toward the SEM model created in this work, with particular focus on discussing human worker job security. This distinct interview portion focused on the initial reactions of experts to the model created, their assessment of human worker job security in AI-enabled workplaces generally, and their expectations for how this model might play out in a real-world AI-enabled workplace.

6.4.2 Qualitative Data Analysis

Just as with Study 2, data analysis for Study 3’s supplemental expert interviews was conducted using in-depth qualitative reflexive thematic analysis [55, 56]. As a reminder, the “reflexive” element of this type of thematic analysis conceptualizes the researcher’s subjectivity as a resource in the process of research, such that coding and interpreting data is recognized as an *“inherently and inescapably subjective practice”* [56].

Following Braun and Clark’s [55, 56] procedures for reflexive thematic analysis, this dissertation’s author first carefully read through the collected data line-by-line to familiarize herself with the data, identifying and notating how these experts in AI-enabled workplaces interpreted Study 3’s quantitative findings and formulated their own views on AI workplace integration that is considerate of the factors that

influence perceptions of human worker job security in AI-enabled workplaces. Next, the iterative coding process began with generating initial codes through assigning codes to individual data and then iterating on the codes to eliminate redundancy. Next, codes were categorized into thematic topics related to the research questions, which developed an initial set of themes related to experts' insights on workplace gender dynamics and AI functional design on evaluations and attributions of credit to workers. Themes were further refined to better capture the data in relation to the research questions, and then again to identify the "essence" of each theme [55, 56]. Finally, as with Study 2, the most compelling quotes were selected as examples to craft a narrative structure for Study 3's findings that flows logically and naturally [56].

6.4.3 Researcher Positionality

While 5.3.3 in Study 2 already details my positionality as a researcher in terms of these interviews, it is still valuable to reiterate here how said positionality impacted how I approached the supplemental expert interviews conducted in both Studies 2 and 3, as this transparency is a vital component of reflexivity [56].

First, it is important to reiterate that my experiences with workplace gender dynamics and job security within and outside of STEM-related fields are rooted in my personal identity as a White, American woman in academia. Three out of four of the women participants are Asian and might therefore experience or interpret workplace gender dynamics differently because of their intersectional identities. However, as mentioned in Study 2, our shared identities as women in academic and industry fields related to AI and tech does allow me to be aware of the unique challenges that women face in U.S. markets in terms of job security, which all participants operate within.

Second, it is important to acknowledge again that all 6 participants are individuals that I knew prior to conducting these interviews, therefore there is a possibility that their insights might have been influenced by their pre-existing perceptions of me and my work. However, their insights as experts are still highly valuable given their well-established years of experience and specific expertise in people management and AI workplace integration (see Table 5.2), and I already had established the kind of rapport and shared understanding with my interviewees that is often valued in rich qualitative work [446].

6.5 Results Part I: Structural Equation Model

6.5.1 Model Fit

The final SEM created for this work is depicted in Figure 6.2. The model created yielded a good overall fit ($\chi^2(311) = 466.24$, $CFI = .996$, $TLI = .996$, $RMSEA = .027$, $90\%CI : [.022 - .032]$). Based on the evaluation metrics put forth by Hu and Bentler ($CFI > .96$, $TLI > .95$, $RMSEA < .05$, and upper $RMSEASI < .10$), this model presents an excellent fit for the data examined. The path model displayed in Figure 6.2 was generated using standardized coefficients, meaning differences in scale type are standardized across the measures explored within the model. Paths between two dependent variable constructs depict standardized changes from one construct to another, and paths between independent variables to dependent variables present standardized changes from one condition to another.

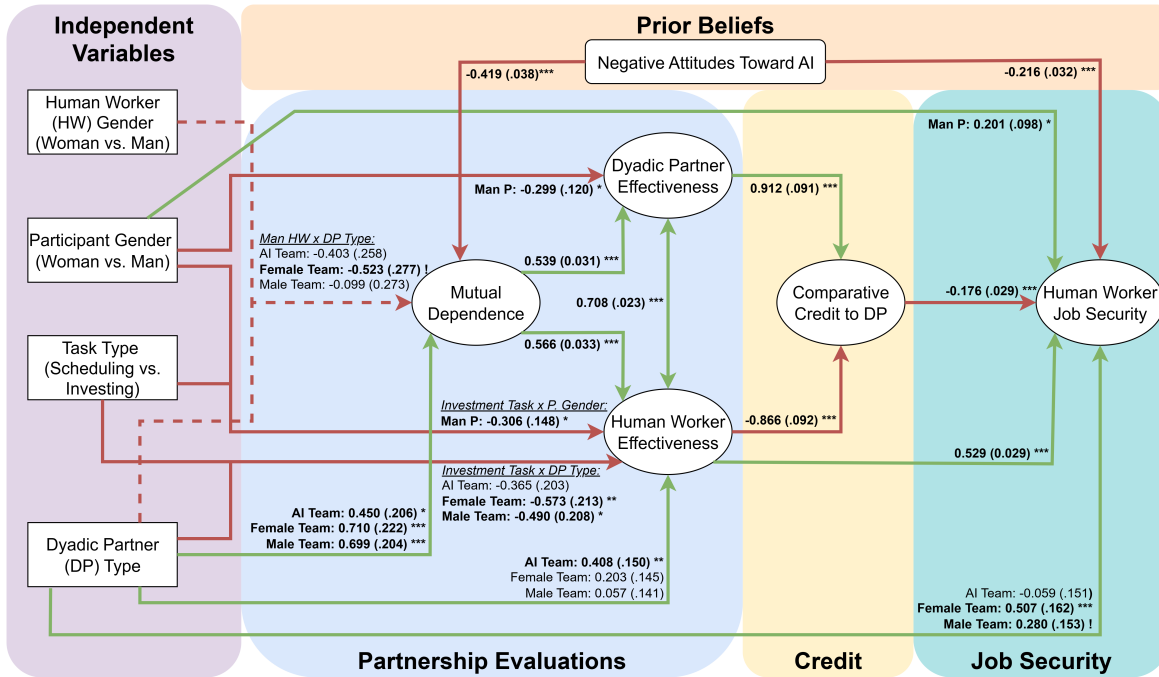


Figure 6.2: The full structural model for AI-enabled workplace partnership evaluations, comparative credit, and human worker job security constructs. Significant results are indicated as follows: * $p < .05$, ** $p < .01$, *** $p < .001$. Each significant result for the partnership evaluation constructs and the job security construct is accompanied by the β coefficients, which signify the standardized increase or decrease in the construct as it corresponds to the increase or decrease of the other construct connected on the path. For the construct “Comparative Credit to DP”, a negative β coefficient indicates that more comparative credit is being given to the Human Worker (HW), while a positive β coefficient indicates more comparative credit is being given to the Dyadic Partner (DP). Furthermore, the β coefficients for the constructs that are contained within the “Independent Variable” block represent the standardized difference between the experimental conditions for that independent variable, and therefore effectively represent effect sizes as measured by Cohen’s d . The number next to each β coefficient in parentheses is the standard error.

6.5.2 Model Examination

Starting with the right-most part of the model created (i.e., blue area labeled “*Job Security*” in Figure 6.2), a multitude of factors are shown to influence how evaluators perceived the job security of the human worker. Starting with direct effects from manipulations (i.e., purple area labeled “*Independent Variables*” in Figure 6.2), changes in the gender of an evaluator (“*Participant Gender*”) from Woman to Man Participant (Woman $\rightarrow$ Man) was shown to significantly improve evaluator perceptions of human worker (HW) job security. This means that men evaluators perceived the human worker as having more job security than what women evaluators perceived. Additionally, changes in Dyadic Partner (DP) Type from AI Tool to Human Teammates (Male and Female) (AI Tool DP $\rightarrow$ Male and Female Teammate DP) was shown to significantly improve perceptions of the human worker’s job security.

In other words, human workers who worked with another human teammate, regardless of gender, were perceived as having more job security than human workers who worked with an AI tool. Stemming from other dependent variables in the model, Human Worker (HW) effectiveness and credit allocated to the dyadic partner compared to the human worker were both shown to significantly influence human worker job security, with the former having a positive effect and the latter having a negative effect. Lastly, participants’ negative attitudes toward AI were shown to negatively influence perceptions of human worker job security. Based on these effects, clear connections can be seen where the evaluation of human workers, their comparison to their dyadic partner, the type of dyadic partner humans work with, and the attitudes toward AI evaluators have all serve to influence the resultant perception of the human worker’s job security.

Moving left within the model, as mentioned in the methodology of this work,

different credit measures were leveraged to create multiple different models. Comparative credit presented the most robust factor explored within the model in relation to model performance and convergence. Perceived comparative credit was shown to be significantly influenced by both the human worker effectiveness and dyadic partner effectiveness evaluations. As expected, increases in human worker effectiveness were shown to decrease credit allocated to the dyadic partner, and increases in dyadic partner effectiveness were shown to increase credit allocated to the dyadic partner. Seeing as this is a comparative measure, where a decrease in credit allocation to one partner means that there is an increase in credit allocation to the other partner, these results can also be interpreted to mean that an increase in human worker effectiveness was shown to increase credit allocated to the human worker, while an increase in dyadic partner effectiveness was shown to decrease credit allocation to the human worker.

Continuing to move left in the model, the effectiveness of both the dyadic partner and the human worker showed a number of direct influences from manipulations as well as other factors within the model. Starting with the simpler of the two, dyadic partner effectiveness was shown to be significantly influenced by the gender of a participant and the mutual dependence perceived between the dyadic pair. Greater levels of perceived mutual dependence increased perceptions of dyadic partner effectiveness, and men participants, compared to women participants, were shown to perceive dyadic partners are being less effective. Shifting to human worker effectiveness, there was a significant main effect from dyadic partner type, with the AI teammate, compared to the AI tool, yielding more positive evaluations of human worker effectiveness.

Additionally, an interaction effect was shown between task type and dyadic partner type, where the female and male human teammates show different impacts on human worker effectiveness based on the task presented. Another interaction effect

was shown between task type and participant gender, where differences between tasks were found based on differences in participant gender. The mutual dependence of the dyadic pair was also shown to significantly influence human worker effectiveness, with increases in mutual dependence leading to increases in human worker effectiveness. Lastly, there was also a bidirectional effect between human worker effectiveness and dyadic partner effectiveness, meaning both perceptions are highly correlated with with the other, such that they positively influence one another.

Moving to the final left-most factor in the model, perceived mutual dependence was also shown to have direct effects from manipulations and covariates. A significant main effect was found from dyadic partner type, with partnerships containing any of the teammate dyadic partners being perceived to be more mutually dependent than partnerships with the AI tool. Additionally, a borderline significant interaction effect was found between human worker gender and dyadic partner type, with the female teammate dyadic partner showing near significant differences between human worker gender conditions. Lastly, participants' negative attitudes toward AI was shown to significantly and negatively influence the mutual dependence perceived between dyadic partners.

6.5.3 Summary of Results and Effects

Based on the effects modeled in Figure 6.2, total effects for manipulations were calculated and graphed to help summarize the results of this model. Main effects of manipulations are shown in Figure 6.3 and interaction effects are shown in Figure 6.4. Based on the effects graphed, similar results can be found between Study 2 and Study 3, where the type of dyadic partner one works with is overwhelmingly impactful to the evaluations of, attributions of credit to, and, ultimately, job security of human

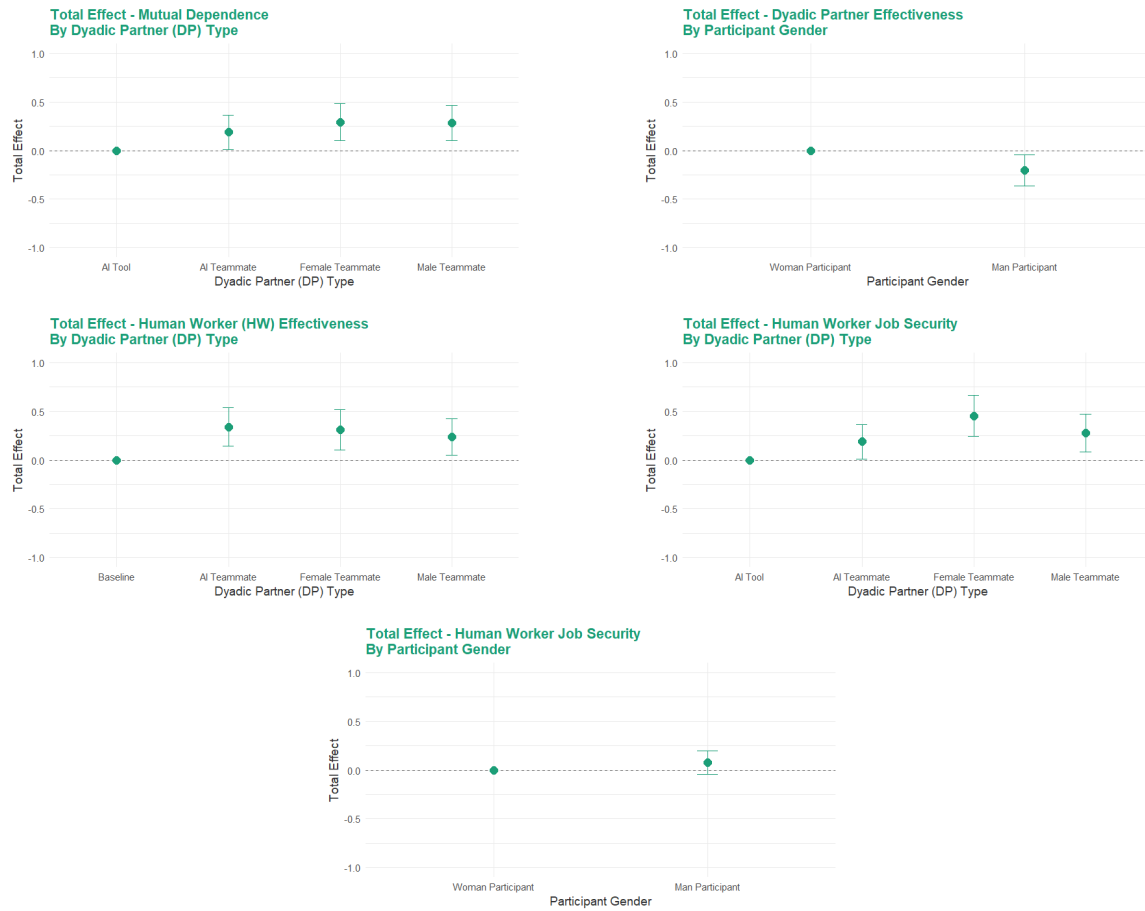


Figure 6.3: Graphical Representation of Total Effects Calculated for the Main Effects of Manipulations in Figure 6.2.

workers. The greatest difference shown within these effects comes from the inclusion of participant gender, as two main effects and one interaction effect was found. For main effects, participant gender was shown to significantly influence dyadic partner effectiveness and job security, and interaction effects showed that the effects of task type were shown to differ based on the gender of the participant.

Shifting away from these specific manipulations and reviewing this model as a whole, the results of this work highlight that the gender and AI dynamics of workers and their partnerships can both directly and indirectly influence the perception that evaluators have of a human worker's job security. Direct effects generally come from

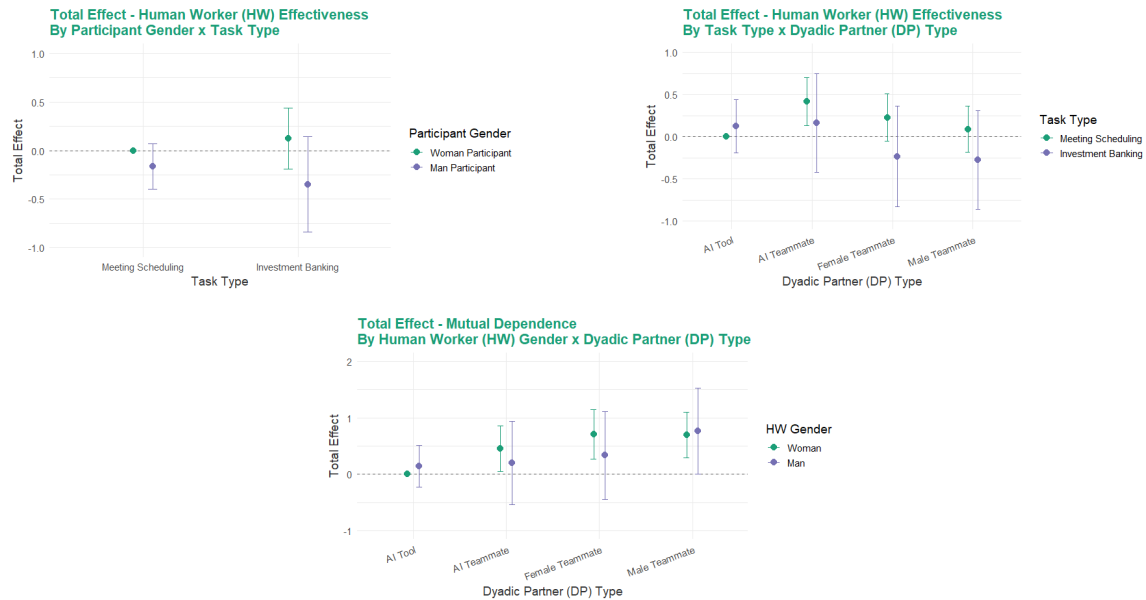


Figure 6.4: Graphical Representation of Total Effects Calculated for the Interaction Effects of Manipulations in Figure 6.2.

the type of dyadic partner humans find themselves working with, but the effects are also mediated by various other evaluations, such as mutual dependence, worker effectiveness, and credit attribution. The result of this iterative modeling process thus yielded a clear and interconnected pathway through which the dynamics of a human worker's workplace can ultimately influence their mutual dependence and effectiveness with their partnership, which, in turn, influences the credit they are attributed within their workplace, which all ultimately serve to determine the level of job security they would expect and others would expect for them in said workplaces. Levering this knowledge, a greater understanding of job security within workplaces facing AI integration can be achieved.

Based on these insights, Table 6.2 and Table 6.3 summarize how hypotheses for S3.RQ1 and S3.RQ2 are supported, partially supported, and not supported. Specifically, for S3.RQ1, results for Study 3 did show that working with a DP that acts as a Teammate significantly improved evaluations of HW effectiveness (**H1 sup-**

ported), and that evaluations of mutual dependence mediated the relationship between both HW and DP effectiveness and the type of DP featured in the partner (**H7 supported**). Additionally, when the partnership was working on the Investment Banking Task, working with the Male Teammate DP, but not the Female Teammate DP, significantly worsened perceptions of HW effectiveness compared to the Meeting Scheduling Task (**H2 partially supported**). All other hypotheses associated with S.3.RQ1 were not supported, as there were no significant differences in evaluations of the Women Human Worker (**H3 not supported**), evaluations of DP effectiveness were not significantly influenced by the type of DP (**H4 not supported**) or evaluators' negative attitudes towards AI (**H5 not supported**), and working with a human DP did not improve perceptions of partnership mutual dependence (**H6 not supported**).

For S.3.RQ2, Study 3's results did demonstrate that evaluations of HW effectiveness do mediate the relationship between comparative credit and the type of DP the HW worked with (**H8 supported**), but DP effectiveness does *not* mediate the relationship between comparative credit and DP type (**H9 not supported**). Moreover, the model shows that evaluations of comparative credit mediate the relationship between perceptions of HW job security and partnership evaluations (**H10 supported**). In addition to these mediation effects, the model does demonstrate that evaluations of HW effectiveness directly significantly improves perceptions of HW job security (**H11 supported**), and that working with a DP that is a human Teammate (Female or Male) does significantly directly improve perceptions of HW job security compared to when working with an AI (Tool or Teammate) (**H12 supported**). However, when working with human DPs, working on the Investment Banking Task significantly worsened perceptions of HW job security, but only when mediated by HW effectiveness (**H13 partially supported**). Finally, Study 3's model does demon-

strate that evaluators' negative attitudes towards AI significantly worsen perceptions of HW job security (**H14 supported**).

H#	Hypothesis	Result
H1	Working with a dyadic partner (DP) that acts as a Teammate (AI, Female, or Male) will significantly improve evaluations of the human worker's effectiveness.	Supported
H2	When the partnership is working on an Investment Banking Task, working with a dyadic partner (DP) that is a human Teammate (Female or Male) will significantly worsen evaluations of the human worker's effectiveness compared to when the partnership is working on a Meeting Scheduling Task.	Partially Supported - (only Male Teammate worsened evaluations)
H3	When the human worker (HW) is a Woman, working with a dyadic partner (DP) that is an AI Tool will significantly worsen evaluations of the Woman HW's effectiveness compared to when she works with any of the dyadic partners that are Teammates (AI, Female, Male).	Not Supported
H4	Having a dyadic partner (DP) that acts in a Teammate (AI, Female, or Male) capacity in the partnership will significantly improve evaluations of the dyadic partner's effectiveness.	Not Supported
H5	The more negative attitudes towards AI an evaluator holds, the more they will perceive the effectiveness of an AI (Tool or Teammate) dyadic partner (DP) to be significantly lower when compared to evaluators with less negative attitudes towards AI.	Not Supported
H6	Working with a dyadic partner (DP) that is a human Teammate (Male) will significantly improve perceptions of partnership mutual dependence compared to when the dyadic partner (DP) is an AI (Tool or Teammate).	Not Supported
H7	Evaluations of partnership mutual dependence will mediate the relationship between dyadic partner (DP) effectiveness, human worker (HW) effectiveness, and the type of dyadic partner (DP) featured in the partnership.	Supported

Table 6.2: Results of Hypotheses for **S.3.RQ1: *What are the empirical links between workplace gender and technological dynamics, evaluator biases, and partnership evaluations in AI-enabled workplaces?***

H#	Hypothesis	Result
H8	Evaluations of human worker effectiveness will mediate the relationship between comparative credit and the type of dyadic partner (DP) the human worker works with to complete the task.	Supported
H9	Evaluations of dyadic partner (DP) effectiveness will mediate the relationship between comparative credit and the type of dyadic partner (DP) featured in the partnership.	Not Supported
H10	Evaluations of comparative credit will mediate the relationship between perceptions of the human worker's job security and partnership evaluations.	Supported
H11	As evaluators' perceptions of human worker (HW) effectiveness increase, perceptions of the human worker's job security will significantly improve	Supported
H12	Working with a dyadic partner (DP) that is a human Teammate (Female or Male) will significantly improve perceptions of the human worker's (HW's) job security compared to when the dyadic partner (DP) is an AI (Tool or Teammate).	Supported
H13	When working with a dyadic partner (DP) that is a human Teammate (Female or Male), working on the Investment Banking Task will significantly worsen perceptions of the human worker's (HW's) job security compared to working on the Meeting Scheduling Task.	Partially Supported - (Mediated by human worker effectiveness perceptions)
H14	Evaluators' biases against the use of AI in society (i.e., negative attitudes towards AI) will be associated with significantly worse perceptions of the human worker's job security.	Supported

Table 6.3: Results of Hypotheses for **S.3.RQ2: *How do evaluations of human worker effectiveness, dyadic partner effectiveness, and comparative credit meld to form expectations of job security for human workers in AI-enabled workplaces?***

6.6 Results Part II: Supplemental Expert Interviews

Results of the supplementary expert interviews reveal how perceptions of human worker job security in the age of AI-enabled workplaces have evolved to reward collaborations between humans and AI, and how workplace gender dynamics shape who has access to these rewards. These results thus shed light on the “why” element of Study 3’s SEM results, specifically by exploring how industry narratives have evolved to view AI as replacing jobs but not workers in a workplace, adding nuance to the surprisingly optimistic results regarding AI teammates’ links to job security through evaluations and attributions of credit (**S.3.RQ1** and **S.3.RQ2**), and (2) providing nuance to Study 3’s results concerning participant gender by highlighting how women workers may still be disproportionately disadvantaged by this shift compared to their more confident male counterparts because of ongoing burdens placed on women by unequal workplace gender dynamics, particularly in STEM-related fields.

6.6.1 Theme 1: AI Can Replace Jobs, But It Cannot Replace Workers.

The first theme from this data focuses squarely on the dynamic landscape of job security in AI-enabled workplaces. As continuously noted by Experts 4 (59, Woman, Asian, CEO of Science/Tech Accelerator) and 6 (28, Woman, Asian, AI Governance Advisor), the use of AI in workplaces is being used to supplement or assist human workers, not replace workers, such that human worker job security should not be significantly impacted by AI workplace integration. However, there still exists a public perception that AI is actively threatening human worker job security,

which could make it more difficult to perform healthy and effective AI integration if not properly accounted for.

Expert 6 particularly emphasizes to the companies she advises that AI integrators have to focus on adaptations and augmentation, not replacement, if they hope to make any traction with AI integration into their workplace. In fact, Expert 6 explains that industry leaders in AI governance and integration are increasingly turning towards how agentic AI can work to collaborate with human workers to take on the kinds of menial but time-consuming tasks that lead to burnout in humans, with the intention of freeing up said humans for more complex tasks. The following quote from Expert 4 builds on this concept by describing her ideal scenario when it comes to balancing AI workplace integration with human worker security through human-AI collaboration:

“If you’re an independent person that’s working on a project, [as CEO] I will encourage to use AI for those tasks that are time consuming, but still actually fact check certain things, but using the tool or a teammate to create an efficiency for those particular tasks. But I also want to encourage a human to human partnership, because I don’t think I will ever consider AI to be 100% correct. Yeah, so I would still have a dyadic type relationship, but each one of those individuals will have their own tool that they can use to do their work at the same time, still have the human component to actually ensure that the information that’s being provided is correct... Originally, I was going to say that maybe an AI teammate that’s specific to the triad (human-human-AI)... It is two different roles, I would say, between a tool and a teammate.”

Building upon this, Expert 6 noted that AI, especially in its current form, is

not going to be able to replace a human worker in their entirety, but rather will replace specific job responsibilities to free up human workers. Experts 2 (29, Woman, White, AI Research & Development), 5 (32, Man, White, Dir. of Business Intelligence), and 6 noted that the current capabilities of AI technology make it much more apt to take over a single responsibility as opposed to an entire worker. These thoughts were, in part, due to the experts' belief that humans make unique contributions to the work they do that cannot be adequately accomplished by AI, such as bringing a level of creativity, emotionality, and generalized intelligence to their work. Moreover, Expert 6 noted that, while agentic AI technology is beginning to achieve a more worker-like and generalized capability, the role of workers can and should still be defined and valued, as the diversity of thought contributed by humans is critical to workplace effectiveness.

When discussing this aspect of defining human worker value, Participant 5 deepened insights by describing how this definition is likely to be domain and role-specific. Expert 6 noted that, in reality, some job responsibilities will be more straightforward to translate to AI. However, many job responsibilities, especially those in high-risk domains such as healthcare, are going to present complex challenges because of the nature of the work itself and a lack of trust in AI's abilities to do such critical work. Even if AI's capabilities progress, Expert 5 noted that these workplaces are still going to need to carve out and define clear procedures for AI and humans, similar to how human-only teams have seen specific role delineation in these contexts. This will be particularly vital in situations where the risk of failure comes with higher consequences. The following is a quote from Expert 5 that expresses this idea:

“You should keep AI out of your core systems, because that’s got the highest risk of screwing something up. Then you have your support systems

and supporting business processes. You could use AI in those, but you need to understand the impact on the core systems business. Then you have analytics and business intelligence, like, we just use AI for that all day... [AI] allows me to analyze more data all at one time, so I can have a quicker decision support tool for a leader. So that's like the least risk, right? No business process is harmed in the making of that, like, people still get paid. But I should not have AI, like, in the middle of my payroll processing, because what if it messes up? That's high risk, and it's going to impact people's lives, so I should not put AI there."

Regardless of the domain, however, this process is going to require a substantial shift in how organizations operate and manage their human workforce. All six experts noted that organization must plan for and integrate AI in a defined and clear manner, with many of the experts providing direct advice for how to approach AI workplace integration while keeping human worker job security in consideration.

First, P6 stressed the fundamental need to create representative ethics boards surrounding AI integration within the organizations that she advises. She especially emphasized that these boards must contain a person who represents every type of person within in the organization, e.g., make sure there are representatives of every gender, ethnicity, etc. present in the workforce on the board. P6 explains that these ethics boards provides a better diversity of thought through representation, and act as a base-level preventative measure to ensure that biases and marginalization are not perpetuated by AI workplace integration design and evaluation.

Second, Expert 5 noted that organizations are going to need some type of external venue to observe and understand AI integration to reduce risks to their own organization, as the following quote illustrates:

“We, as an organization, because we’re a nonprofit, we don’t have as many resources. We should not be using AI to try to keep us on the cutting edge of like, profitability, right? So therefore, we should let other people screw it up first, figure out where it’s best used, and where it’s not quite a good use, or how we actually engage with AI, and then [integrate AI] as we get lower stakes opportunities to engage with AI.”

Lastly, Expert 4 noted that organizations must begin searching for new opportunities to leverage AI that extends human expertise and knowledge via AI, as the following quote illustrates:

“So I’ll give you an example. For the most part, our team is not as experienced in the FDA regulation in the nutraceutical space, but we’re very experienced in the drug therapy, life science space, where it (regulation) requires rigorous FDA compliance. Regulation in the nutraceutical space is not as rigorous, and we have some knowledge, but our expertise is not as deep as the other. So if we are working with a company and they’re in the nutraceutical space, we need to understand how we would carve out the proposal [we give them] for engaging us in that process for FDA approval in the nutraceutical space. So that’s when we asked AI to give us information, ‘What is the current standard and regulation for FDA approval in the news about over-the-counter nutraceuticals?’ And we took that information, we plugged it into our proposal for the process.”

Given the expert opinions provided, some clear summarizations and takeaways can be gleaned from this work. First, while human worker job security may not be immediately threatened by AI, the perception and expectation of this threat will make integration harder if not properly considered. In turn, it is on organizations

to bear the burden of understanding when and how to integrate AI, with the goal of augmenting and empowering workers rather than replacing them. However, doing so is not only going to require substantial organization-wide effort, but it will also require domain-specific considerations that account for specific factors that contribute to human job security and roles within a context, such as the gender makeup of the workforce.

6.6.2 Theme 2: Women May See Greater Personal Burdens to Overcome Gender Biases in AI-enabled Workplaces

Shifting toward the workplace gender dynamics featured in the model created, multiple experts immediately honed in on how and why participant gender directly influenced human worker job security and dyadic partner effectiveness. More specifically, when asked why they thought that the men participants rated human worker job security higher than women participants did, Experts 2, 3, and 4 all had the exact same immediate response of “*Confidence*,” meaning they believed the difference to be a result of men’s overconfidence in their abilities, such that they might feel they are more capable than their partners. In turn, male evaluators may see a greater level of (over)confidence in the job security of human workers in AI-enabled workplaces. In turn, it may be important to further investigate and possibly mitigate the (under)confidence that women might have in these same positions. Expert 3’s recommendations for women workers going forward in light of Study 3’s results further discusses this topic:

Researcher: “What recommendations might you give individuals, especially women who might be facing impending AI integration into their workplace, given all of these results we talked about?”

Expert 3: “Leverage AI as their ally, both through its functions to help with their daily work and also as an ally to help build up their confidence. And yeah, through, you know, giving her facts about, ‘No, men are just being confident, they can just brag about what they are capable of, and you can do that too.’”

Interestingly, Expert 5, who is a man, had a different perspective as to why male participants have different evaluations and job security perceptions. In particular, they noted that some differences between men and women evaluators might have something to do with anxiety, meaning that men might repress their anxiety about job insecurity while women might be more open about expressing anxieties about job security. In this way, Expert 5’s sentiments also speak to overcompensating, but from the perspective that men might be overcompensating to hide their anxieties. The following quote demonstrates Expert 5’s musings after being shown the model for participant gender:

“I guess it’s hard to tell... Did you measure for psychological anxiety?... Maybe... Guys are probably less likely to be seeking help for mental health issues, even if they have them, while women may be getting treated, like women are more likely to be treated for mental health things... So maybe, I don’t know, maybe there’s something there?”

Regardless of why this effect may have existed, a few experts also built on the results of this work to highlight how these impacts may ultimately create more personal burdens on women. Specifically, Expert 2 noted that, while it may be unfair to women, this continues to highlight the need for women to develop strong technical literacy and skills in the workplace. On the one hand, these skills would provide them with the ability to meet any challenge forth. On the other hand, it would also help

improve their confidence in a workplace constantly being changing by AI technology. Alternatively, Expert 6 had a different perspective, as they felt these gender biases were going to be less pronounced in actual office workplaces; however, they did note that many of these biases may themselves be perpetuated by biased and sexist AI technologies. Expert 6 saw women having a greater personal burden placed upon them in the use of AI technology as they may have to overcome using a tool that presents some bias and sexism toward them.

For both of these examples, experts were clear that while this is not the fault nor should it be the responsibility of women workers, it is unfortunately going to burden them, but they can do something about this burden. Reflecting on a recent experience she had with a male colleague berating her about her AI knowledge, the following quote from Expert 2's elaborates on what she does to combat these attitudes in AI workplaces:

“I think there's just going to be those folks that, if your job is AI related, are still going to question you [because you are a woman], and they tend to be men, at least in my organization. The best way I have to combat that has been to show my knowledge and show that I do have the technical chops... And so continuing to work with that person feels like I just have to keep proving myself, which kind of goes back to the point I was making of you [as a woman] kind of have to prove yourself... You don't want to seem like the weak link of that partnership, or like, especially to leadership, if you're being expected to use AI, you want to be able to hit the ground running, because you don't want to be the person left behind.”

Synthesizing this theme, Study 3's model highlights that while gender dynamics do not play a substantial role in human worker job security, that role is

present. Expert opinions highlight that these gender biases are likely going to influence job security, often through mediating factors such as effectiveness or credit. However, despite these effects being small, their impacts were likely going to require women to personally recognize and sometimes overcome differences in their experience in AI-enabled workplaces. While some women workers may have an accommodating evaluator that is away of these issues, many are likely going to need to personally bolster their confidence and capability surrounding the use of AI technology while still addressing its flaws. It should be the responsibility of researchers to assist these workers in this endeavor.

6.7 Discussion

6.7.1 Understanding Job Security in the Age of AI

The ultimate goal of Study 3 was to create a model that connects workplace gender dynamics, sociotechnical beliefs, worker effectiveness and credit to better understand how AI-enabled workplaces will shape perceptions of human worker job security. Based on the model created, a clear path towards perceptions of job security emerges, where (1) the gender and technological dynamics of a workplace can influence how evaluators perceive the partnership's mutual dependence and the perception of effectiveness of human *and* AI worker evaluations, which can, in turn, (2) influence attributions of credit between partners, and, ultimately, (3) perceived expectations of human worker job security.

Through this complex modeling, Study 3 underscores how the way that evaluators and, by extension, workers themselves, make sense of their job security in AI-enabled workplaces will be intimately intertwined with the workplace gender and

technological dynamics of their workplace, not just whether AI is integrated or not. Indeed, based solely on the model presented, one can see that the workplace task context the partnership completes, the type of AI (tool v. teammate) humans work with as part of that partnership, and even the gender identity of the person evaluating the partnership all play complex roles in the perceived job security human workers will have in the coming age of AI-enabled workplaces.

Within the HCI domain, this shift toward understanding, investigating, and representing job security as one component of a more complex workplace dynamic-based narrative presents a substantial departure from traditional HCI perspectives on AI workplace integration.

On the one hand, prior work in HCI and AI research has been more heavily focused on the concept of acceptance of AI workplace integration [483], with the goal of shaping human workers such that they understand and want to work with AI technologies [303]. When translating these concepts to job security, research often focuses on how human worker acceptance of AI influences perceptions of job security [111], implying that job security concerns among workers stem almost exclusively from lapses in AI literacy and, thus, acceptance among these workers. Moreover, even when research presents a more complex and sociotechnical view of how perceptions of job security in AI-enabled workplaces comes about, it often focuses on meta- and society-level factors, positioning AI technology as having a monolithic impact on job security across the board [111].

On the other hand, the recent reports (October 2025) that have emerged demonstrating that the job market in the U.S. has actually *not* been significantly impacted by AI workplace integration [182, 482] provide a clue that human worker job security concerns in the age of AI requires more nuanced and targeted investigation that goes beyond blanket statements and assumptions about AI's impacts. Study 3

thus significantly deepens the nuances that underlay these prior narratives by testing and subsequently modeling multiple pre-existing factors within a workplace that are likely to influence expectations of job security, including workplace gender dynamics, AI functional design, and workplace context-specific task assignment.

Within the Human-AI Teaming (HAT) domain specifically, understanding this model of job security will be pivotal to translating the concept of human-AI teaming into real-world teams. Within both Study 2 and Study 3, the dyadic partner with whom one works was shown to be the most impactful driver of how evaluators perceived the human worker's job security given their existence within an AI-enabled workplace. Building on prior work in human-AI teaming, this work partially contributes to the knowledge that AI teammates have the potential to harm perceived job security [156, 207]. Yet, the interconnected relationship between perceived human worker effectiveness, which was actually shown to improve when working with an AI Teammate, and perceptions of that human worker's job security demonstrate that AI teammates are not *universally* detrimental to how others view a worker's job security.

Indeed, emphasizing the effectiveness and mutual dependence gained when designing AI workplace integration to include AI teammates may even present an opportunity for human-AI collaboration via teaming to *counteract* the negative effects that AI might otherwise have on perceptions of worker job security by improving perceived worker effectiveness, which, in turn, significantly improves job security perceptions. This has significant implications for teams, as job security has been shown to be a substantial influence on human performance within teams [261]. As human-AI teams operating in everyday workplaces continues to become more of a near-future reality thanks to the rapid advancements in AI technology, confronting these job security concerns head-on will be fundamental to preserving team and individual performance in AI-enabled workplaces.

As such, to work towards ensuring that the success of AI-enabled workplaces in real-world industries is not significantly hindered by human worker job security concerns, future research should work to leverage and expand the model provided by this dissertation's Study 3 in three ways.

First, human-AI teams need to better measure and understand the workplace gender dynamics and workplace context-specific task complexities present in organizations, as doing so will help inform their ability to model job security within their team in a manner that is more tailored to their specific workplace functions and culture. Second, any workplace contemplating the integration of AI technology needs to set a clear outline of how worker effectiveness and attributions of credit are determined in the worker evaluation process, and how these evaluation factors are influenced by the use of an AI teammate so that adjustments can be made to account for and reward that usage. Lastly, future research needs to expand upon the factors that might influence perceived job security in AI-enabled workplaces, as other social (i.e., race, age, more complex gender identities) or technological (i.e., AI usage levels, AI capabilities, workplace training for AI use) factors may continue to shape how AI impacts the future workforce in new and complex ways.

6.7.2 Design Recommendations for Approaching Future AI Workplace Integration with Consideration of Human Workers' Job Security

Grounded in both the quantitative and qualitative results of Study 3, the remainder of this chapter presents two foundational design recommendations aimed at encouraging organizations to approach AI workplace integration with consideration for how this approach will influence their human workers' perceptions of their own

job security, particularly in the context of women workers. It does so by first explicitly calling on organizations to create task-specific definitions for how credit will be attributed to human workers versus AI workers, promoting transparency and keeping their human workforce informed. This section then details how organizations should take responsibility for preparing their workforce for the potential impacts that AI integration might have on worker job security *prior to* any integration efforts, namely through the formation of ethics and innovation committees that address job security through proactive consideration of all employees and their needs.

6.7.2.1 Workplaces Need to Create Task-specific Definitions for How Credit Will Be Attributed to Humans, AI Tools, & AI Teammates

One of the insights to come from both the model created and expert's reactions to this model was the need to have a clearly defined criteria for credit as it influences job security. Experts noted that while effectiveness, credit, and job security all interrelated, the influence of AI on these factors may only be present when completing tasks that directly rely on AI assistance. This insight may also serve to highlight task type's role within this model. As such, pulling from these insights, the first design recommendation of this work posits that workplaces must create a clear definition of how credit will be determined within each task and if the usage of AI tools or teammates will influence this credit and, ultimately, job security.

On a surface level, this job recommendation predominately focuses on what AI enabled workplaces can do to ensure they better accommodate humans who plan to leverage AI systems. However, this design recommendation also has implications for the design of actual AI technologies. Should a workplace decide if and how AI contribute to the credit attributed to a specific task, it will be fundamental that de-

signers and developers leverage these specifics to design the capabilities and efficiency metrics of AI tools and teammates. In turn, the implementation of AI would better target these credit attributing roles, allowing AI tools and teammates to also develop greater levels of effectiveness, which would subsequently influence job security. Through this dual human and AI focus on credit, future workers will gain greater confidence in how they are expected to leverage AI technology within their specific task and role.

6.7.2.2 Companies Need to Form Representative Ethics and Innovation Committee to Address Job Security, Re-skilling, and AI Innovation Simultaneously Prior to Any AI Integration

One of the two concrete suggestions to come out of the expert interviews was the critical importance for (1) all organizations contemplating the integration of AI to develop an ethics board specifically devoted to evaluating the potential impacts of AI and (2) organizations to identify new opportunities for AI integration that meld human and AI intelligence. This design recommendation merges these two suggestions, with the goal of creating AI ethics and innovation boards. In doing so, the goal of this recommendation is to directly couple any innovation within the AI space with an ethical consideration. In other words, any time a new AI innovation is proposed to an organization, the ethical implications of said integration would already be accounted for. Moreover, as taken from the expert interviews, these boards would need to consist of employees representative of race, gender, and position, both minority and majority representation as well. In doing so, any ethical and innovation decisions are derived from employees, where these eventual innovations will ultimately impact.

Within organizations, these ethics boards are not uncommon [309]; however,

their purposes are generally broad [309]. Shifting toward an AI-specific ethics and innovation board will be critical, as the unique impacts of AI are going to be quick and dynamic. In turn, there needs to be an arm of an organization that is able to more quickly react to these changes, provide clear guidance directly take from employees themselves. This separate board would provide two distinct advantages [355]. First, the representation of this board would ensure that ethical considerations account for gender, racial, economic, and other differences among employees. Second, the inclusion of such a board would provide a place for organizations to strategically identify new opportunities for AI integration and their impacts, with the new opportunities being employee derived from the beginning. By not splitting focus, organizations will see a greater ability to integrate AI effectively and ethically.